

Adilkhan Salkimbayev

Applied Mathematics & Scientific Machine Learning Researcher
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EDUCATION

Kazakh-British Technical University (KBTU)

B.Sc. in Automation and Control | GPA: 3.58 | Aug 2023 – July 2027 (Expected)

- **Relevant Coursework:** Linear Algebra for Engineers, Differential Equations, Calculus of Complex Variables
- **Military Department:** Sergeant of the Reserve (Completed June 2025).

PUBLICATIONS AND PREPRINTS

- **A. Salkimbayev.** "Certifiable Approximation of Model Predictive Control Laws: A Shape-Constrained Polynomial Read-Out." *Major Revision, Journal of Process Control*.
- **A. Salkimbayev, K.S. Haider.** "Robust Adaptive Kolmogorov–Arnold Neural Control." *Under Review, International Journal of Adaptive Control and Signal Processing*.
- **A. Salkimbayev.** "Leray–Hopf Weak Solutions of the 2D Navier–Stokes Equations on the Torus: An Expository Construction." *Expository article; preprint, 34 pp.*
- **A. Salkimbayev.** "Rigidity of the Euler–Arnold Form under Energy-Preserving Deformation." *Paper I of eight; preprint, September 2026.*
- **A. Salkimbayev.** "The Isotropic Directional Obstruction for Cascade Operators." *Paper II of eight; preprint, September 2026.*
- **A. Salkimbayev.** "Cross-Triad Rigidity for Cascade Operators." *Paper III of eight; preprint, September 2026.*
- **A. Salkimbayev.** "Momentum Helicity Separates Cascade Operators from Euler–Arnold Operators." *Paper IV of eight; preprint, September 2026.*
- **A. Salkimbayev.** "A Beale–Kato–Majda Criterion for Right-Invariant H^s Metrics on $\text{SDiff}(T^3)$." *Paper V of eight; preprint, September 2026.*
- **A. Salkimbayev.** "Two-and-a-Half-Dimensional Flows of the H^s Euler–Arnold Equations: A Self-Gravitating Boussinesq System." *Paper VI of eight; preprint, September 2026.*
- **A. Salkimbayev.** "Axisymmetric Flows with Swirl of the H^s Euler–Arnold Equations: A Reduction, Two Thresholds, and a Missing Wall." *Paper VII of eight; preprint, September 2026.*
- **A. Salkimbayev.** "A One-Dimensional Wall Model for the H^s Euler–Arnold Equations: Blowup Below $s = 1/2$, Regularity Above." *Paper VIII of eight; preprint, September 2026.*
- *All eight of the above were produced with substantial AI assistance under my direction; I am responsible for the contents. Not yet independently reviewed.*
- **A. Salkimbayev.** "Eight Conditions on the Missing Estimate for the Three-Dimensional Navier–Stokes Equations." *Note; preprint, September 2026.*
- **A. Salkimbayev.** "Transport and the Supercriticality Gap in 3D Navier–Stokes, I." *Part I of four; preprint, September 2026.*
- **A. Salkimbayev.** "Transport and the Supercriticality Gap in 3D Navier–Stokes, II: The Flow Map." *Part II of four; preprint, September 2026.*
- **A. Salkimbayev.** "Transport and the Supercriticality Gap in 3D Navier–Stokes, III: The Ledger." *Part III of four; preprint, September 2026.*
- **A. Salkimbayev.** "Transport and the Supercriticality Gap in 3D Navier–Stokes, IV: Events." *Part IV of four; preprint, September 2026.*
- **A. Salkimbayev.** "What Must Be Conserved: Thirteen Conditions on the Missing Quantity for the Three-Dimensional Navier–Stokes Equations." *Note; preprint, September 2026.*
- *The six entries from "Eight Conditions" on are a second program, on transport in 3D Navier–Stokes; none resolves the regularity problem. Also produced with substantial AI assistance under my direction; not yet independently reviewed.*

RESEARCH EXPERIENCE

Independent Undergraduate Researcher | Kazakh-British Technical University

Functional-Analytic Control Architecture & PDEs (Current)

- Developing a novel functional-analytic control architecture drawing upon operator theory and methodologies from Brezis, Evans and Stein.
- Translating control frameworks into rigorous mathematical formulations applicable to the study of partial differential equations and fluid dynamics.
- Wrote a 34-page expository construction of Leray–Hopf weak solutions to the 2D Navier–Stokes equations on the torus, proving uniqueness in two dimensions and locating precisely where the three-dimensional estimates fail to close.

Adaptive Control & Nonlinear Dynamical Systems (Jan 2026 – Jul 2026)

- Developed a learning-based adaptive control architecture combining Recursive Least Squares with symbolic Kolmogorov–Arnold Networks (KANs).
- Derived rigorous Lyapunov stability guarantees (ISS / UUB) over the physically realizable operating envelope, with a certificate that requires no persistent excitation.
- Evaluated the algorithm on a quadruple-tank benchmark via a 50-seed Monte-Carlo campaign in Python, demonstrating a branch-free per-step runtime bounded a priori and a 7% aggregate tracking improvement over LTV-MPC.

Constant-Time Explicit Control Laws from MPC (Nov 2025 – Apr 2026)

- Designed a certifiable explicit replacement for Linear Time-Varying MPC on the quadruple-tank benchmark in its non-minimum-phase configuration: an analytic feedforward term, a detuned LQR core, and a learned correction behind a gate that vanishes quadratically at the setpoint, making zero steady-state offset and local exponential stability structural rather than dependent on fit quality.
- Built a two-step symbolic read-out in which a Kolmogorov–Arnold Network (KAN) supplies only the monomial support and the coefficients are re-estimated by a convex quadratic program under affine negative-feedback constraints — eliminating a sign pathology present on up to 30% of the operating box for at most 0.2 percentage points of accuracy.
- Established the constant-time claim structurally in software-in-the-loop — a fixed chain of 30–203 multiply-accumulate operations with no data-dependent branching, stable under $\pm 20\%$ parameter perturbation in 97–100% of Monte-Carlo trials — and reported the negative result that the KAN-selected support is statistically indistinguishable from orthogonal matching pursuit at a matched arithmetic budget.

ADVANCED INDEPENDENT STUDY

Self-directed graduate sequence in analysis and PDE, directed toward the rigorous study of fluid dynamics:

- **Completed:** *Introduction to Real Analysis* (William F. Trench); *Functional Analysis, Sobolev Spaces and Partial Differential Equations* (Haim Brezis), Chs. 1–9 — including semigroup theory (Hille–Yosida) and the variational formulation of elliptic boundary-value problems; *Partial Differential Equations* (Lawrence C. Evans), Chs. 5–6 — Sobolev spaces and second-order elliptic theory.
- **In progress:** *Real Analysis: Measure Theory, Integration, and Hilbert Spaces* (Elias M. Stein & Rami Shakarchi), Chs. 1–3.
- **Planned (Fall 2026):** *Measure Theory and Fine Properties of Functions* (Lawrence C. Evans & Ronald F. Gariepy); *Singular Integrals and Differentiability Properties of Functions* (Elias M. Stein).

TECHNICAL & COMPUTATIONAL SKILLS

Mathematics & Control Systems: Operator theory, partial differential equations, semigroup theory, Lyapunov stability analysis, adaptive control, nonlinear control systems

Scientific Machine Learning: Kolmogorov–Arnold Networks, system identification, embedded SciML.

Programming & Hardware: Python (NumPy, SciPy, matplotlib, pandas, PyTorch), C/C++, STM32 CubeIDE, LaTeX (VSCode).